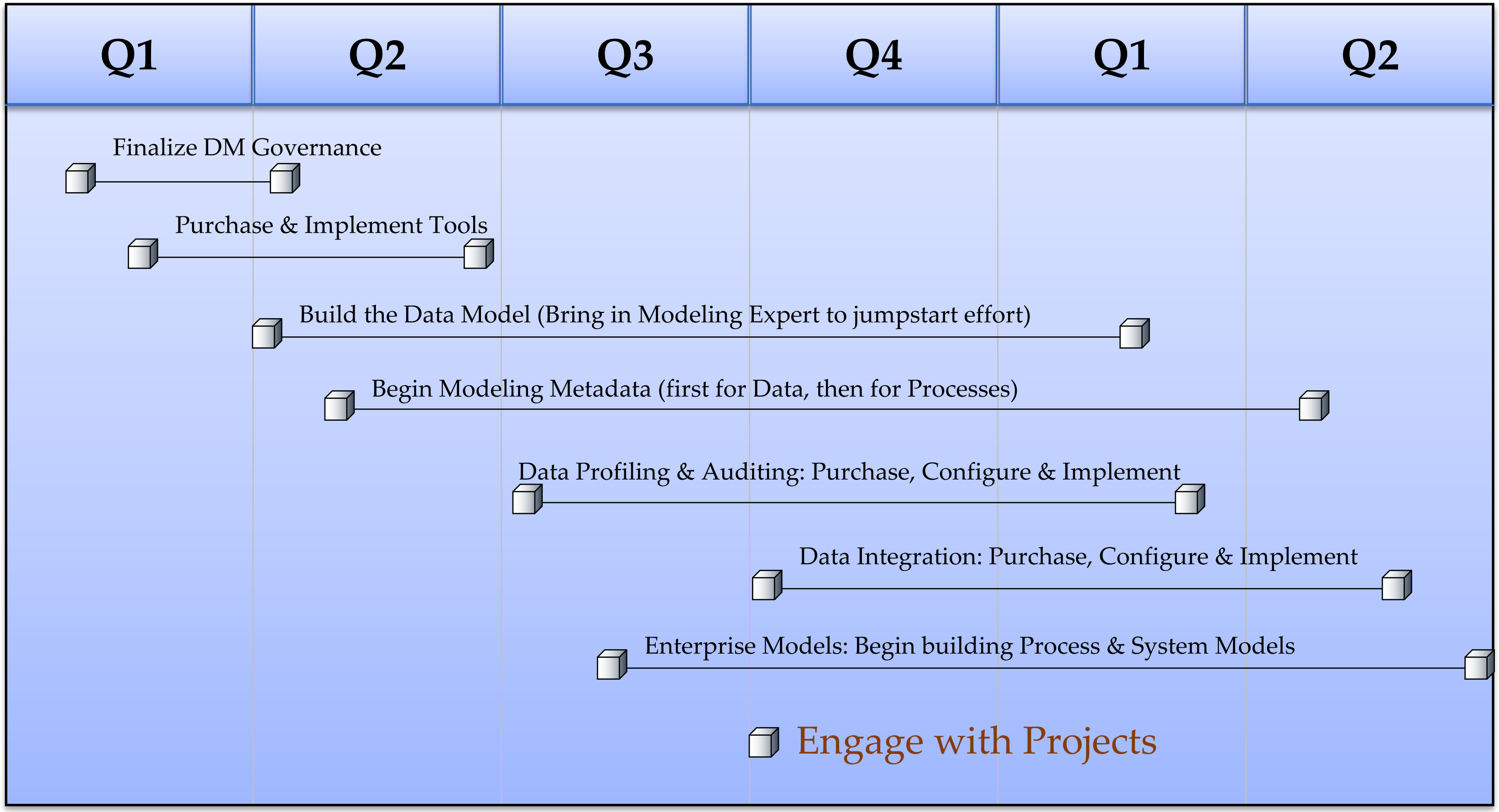
Data Management Roadmap

A progressive approach towards building an Information Architecture strategy

Business and IT Drivers

- Support for business agility and innovation
- Faster time to market
  - Improve capacity and responsiveness of IT
  - Improve operational effectiveness
- Reduce costs – high maintenance costs
- Allow for business partner integration
- Mergers and acquisitions
- Regulatory drivers

High-Level Timeline



Barely reactive to changing business needs

- Lack of business agility and flexibility
  - slow in addressing changing business requirements
- Increasing complexity and little re-use
  - Monolithic applications, tightly coupled rigid systems - very fragile
  - multiple development platforms, tools, and support teams
  - custom point to point interfaces – expensive and complex to maintain
  - Lack of interoperability
- Difficultly in quickly deploying industry standards
- Data quality and data management issues -
  - Multiple, inconsistent sources exist for each data entity (master data)
  - Infrastructure delays due to replication and nightly batch processing
  - Difficulty integrating data in real-time data based on business events

Shift in Strategy: Information Architecture

The IT landscape is changing…

Traditionally	= application-centric, ERP is the centerpiece
New	= Data is the foundation
	= Metadata drives the enterprise
	= Integration depends on data

→ Data Integrity is a core competency

→ Adaptability, Agility and Accountability are key enablers

→ Information Architecture is the cornerstone for DM

→ Formalized Tools, Processes & Governance

Shift in Focus: Model-Driven Approach

• Function oriented • Build to last • long development cycles	• Process oriented • Build to change • Agility & Adoptability
• Application silos • Tightly coupled • Object oriented • Known implementation	• Orchestrated solutions • Loosely coupled • Message oriented • Metadata Driven

Data Modeling (Data, Metadata & Lineage)

Examples of Key Drivers	How can Data Modelling help ?
1. Reduce IT costs **2. Support business agility and growth** **3. Improve Integration** **4. Faster Change Management** **5. Improved Flexibility** **6. Support consistency in data manipulation and transforms**	**Lower workload**: 1. Reduced effort for diagnosing data issues 2. Simplified analysis for identifying dependencies 3. Centralized model for all data **Lower costs**: 1. Fewer resources for manual investigation 2. Integration Planning far simpler 3. Reusable assets (Metadata) **Flexibility**: 1. Easier to plan and manage data changes 2. Changes to data easier to develop/integrate 3. Improved consistency when applying metadata to for data manipulation **Process Automation**: 1. Ability to connect Strategic to Tactical 2. Automation of manual processes 3. Ability to align data across the enterprise

Data Analysis (Profiling, Auditing & Cleansing)

Examples of Key Drivers	How can Data Analysis help ?
1. **Reduce IT costs** 2. **Support business agility and growth** 3. **Improved Data Integrity** 4. **Reliable Supply Chain** 5. **Optimized Data Integration** 6. **Improved Data Quality within and without the enterprise**	**Lower workload**: 1. Pre-configured business content 2. Lower integration effort (apps + partners) 3. Outsourcing of standard processes/services **Lower costs**: 1. Infrastructure 2. Proactive cleansing & profiling reduces rework 3. Applying Business Rules consistently **Flexibility**: 1. Centralized management of applied metadata 2. Ability to integrate metadata into a single set of tools for Profiling, Auditing & Cleansing 3. Easy to adapt metadata to meet business needs **Process Automation**: 1. Ability to improve data quality across each technology layers and organizational boundaries 2. Ability to improve day-to-day operations 3. Ability to improve SAP MFG project

Data Integration Framework (Using Metadata)

Examples of Key Drivers	How can a Data Integration Framework help ?
1. Reduce IT costs **2. Support business agility and growth** **3. Best Integration** **4. Manage Supply Chain** **5. Manage IT outsourcing** **6. Support divisional business models**	**Lower workload**: 1. Pre-configured business content 2. Lower integration effort (apps + partners) 3. Outsourcing of standard processes/services **Lower costs**: 1. Infrastructure 2. Integration 3. Reusable assets **Flexibility**: 1. Less complex applications (less code/bugs) 2. New services easy to develop/integrate 3. Services easy to adapt to business process change **Process Automation**: 1. Workflows across applications 2. Automation of manual processes 3. Process integration with business partner

Modeling requires a Cultural Shift

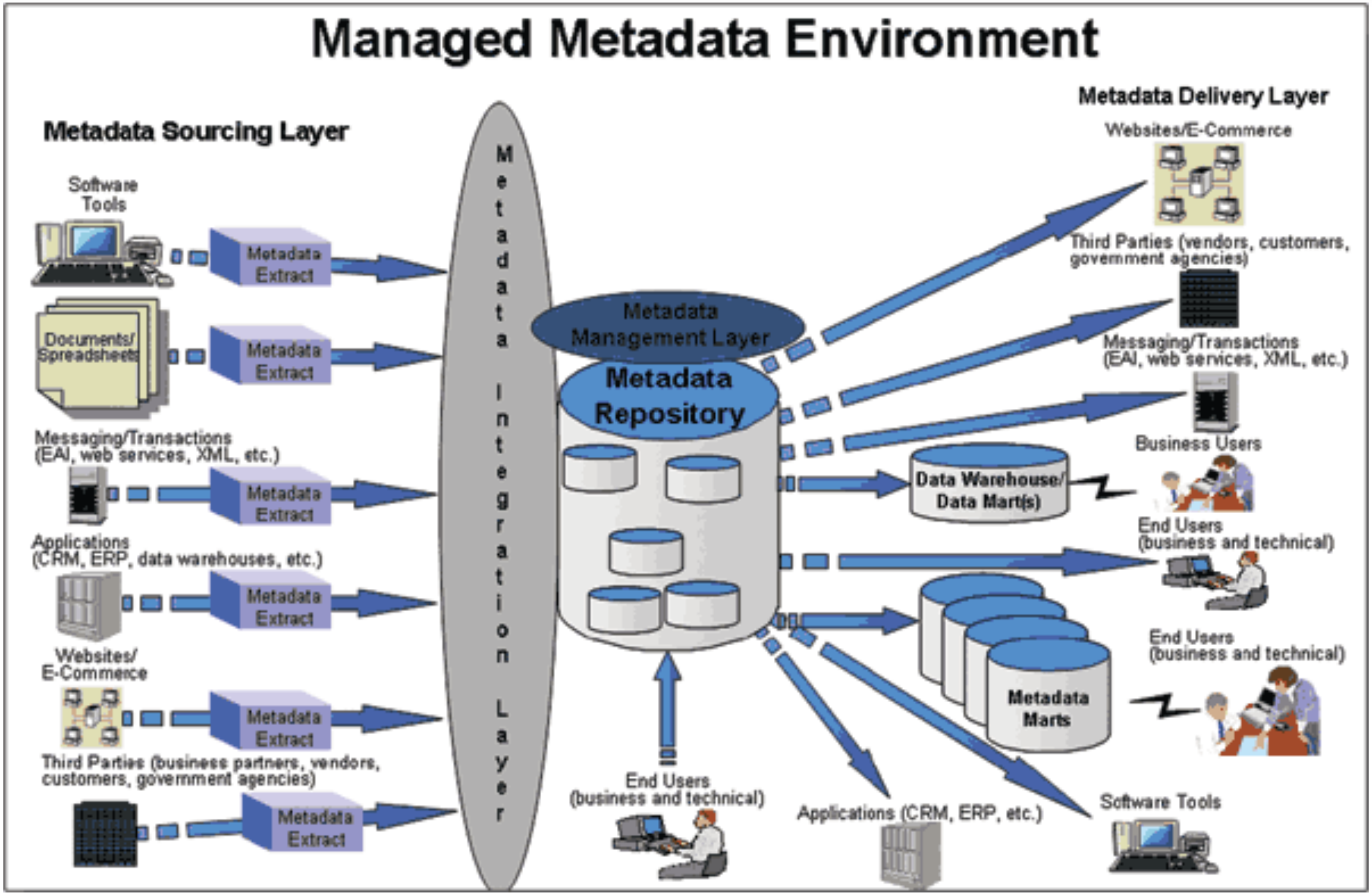
- Migration to MDA – where to start?
- Decomposition of existing rules to derive Metadata
- Begin leveraging Metadata to drive the models
- Implement formal governance to manage change
- Shift the focus to process knowledge and architecture
- Change of attitude and learning new skills (business process analysis, modelling, etc)
- Leverage the models to plan, manage and execute change
- It will take time…years

Leveraging Metadata for Data Harmonization

- Data harmonization is the process of mapping the "current" state of our data (as defined in the data assessment phase) to the target "end" state of our data (as defined in the data alignment phase).
- Managed Metadata is the key foundation in the data harmonization phase. A Metadata repository contains the core enterprise-wide metadata management and utilization.
- The IT staff and data stewards will directly interact with the Metadata repository to perform the process of mapping the current data state to the end data state. *

* This process is vital in many applications as it allows an organization to transform their data into a single version of the truth.

A Metadata Management Environment



- **The goal of the solution implementation phase is to physically build the data models, processes and procedures to achieve the data harmonization. During this phase, several items must be considered.**
  - First, as bad/inaccurate data is identified, it must be tagged and placed into a database for reporting purposes.
  - Second, the business users must be actively involved in the process of examining the inaccurate data so that feedback can be provided back to the data's source.

Solution Implementation

- **The goal of the solution implementation phase is to physically build the data models, processes and procedures to achieve the data harmonization.**

- **During this phase, several items must be considered.**
  - Third, data will error-out because the business rules for the data are not complete or applied consistently. At this point, the business users should be able to identify these situations, so they can:
    - Complete / Define the business rules (Metadata)
    - Update the processes that are (or should be) using Metadata

Solution Implementation

- **The goal of the solution implementation phase is to physically build the data models, processes and procedures to achieve the data harmonization. During this phase, several items must be considered.**
  - Lastly, records that error-out should be updated as new data becomes available or if the source record itself has been updated. In these cases, the records would then recirculate back into the process. In the case of a data warehouse, these previously tagged inaccurate records would then be marked as "clean" and subsequently loaded into the data warehouse.

Continuous Improvement

- **Data assurance is an evolutionary process. Therefore, organizations need to have an ongoing feedback mechanism and update capability to modify and improve its performance.**

- **Meta data and proper meta data management is the key to this initiative as it supports all of the six phases discussed here.**
  - For example, all of the information collected during the monitoring of an ETL process is meta data. This meta data needs to be persistently stored and historically tracked within the Metadata repository so that the IT staff and data stewards can leverage it to accomplish each of the phases of the road map.

Prerequisites For Building an IA Strategy

- ❑ **Assess - understand what is needed to support the business – first**
- ❑ **Build your own Architecture competency organization**
  - Develop your architecture strategy
  - Consider your architectural expertise and applications development skills
  - Consider your development methodology, guidelines and standards
  - Consider your architectural components (infrastructure and tools used)
  - Consider your project management expertise, training and communications
  - Document your business process
  - Information architecture (common Data & Metadata Models)
- ❑ **Build the Information Architecture roadmap - leverage reuse**
- ❑ **Plan for iterative implementation and development - crucial**

Possible Risk Areas

Models are only as good as the foundation that they sit on!

- METADATA is a key to establishing a “single source of the truth”
  - one dictionary breeds understanding, two or more breed confusion!
- Data must be profiled, audited and cleansed to ensure it remains a single source of truth (from inception through downstream usage).
- Data must be standardized and in some cases centralized.
- Data should only land once (where possible).
- Data Models should be used for planning & managing change.
- Data Integration Framework must leverage Metadata.

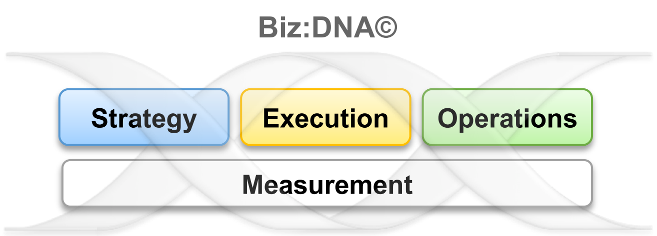
Evolutionary Approach for Building an IA

- Tools and technologies will not automatically give you an Information Architecture
- Architecture without good data is doomed to failure
- Architecture without governance will NOT realize full value

- ❑ Information Architecture is a foundation for modeling the core assets of any business.
- ❑ Modeling is the first step in mapping our information world and planning a sustainable enterprise architecture.
- ❑ Modeling is IT's strategic blueprint towards building our roadmap
- ❑ Long term cultural shift

Thank You

Measure what matters
Create and deliver the right value
Build and nurture a winning culture
Proactively manage change & complexity



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